

GrammarForge - Project Report

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Project Overview

GrammarForge is an AI-powered grammar checking and writing analysis REST API built with OCaml and the Dream web framework. It checks grammar, computes readability scores, analyzes writing style, and suggests text improvements using deterministic analysis algorithms.

Technical Stack

Language: OCaml 5.3.0

Framework: Dream 1.0.0~alpha8

Build Tool: dune + opam

Architecture: REST API with functional analysis pipeline

AI Integration: MockClaudeClient (rule-based grammar/style analysis)

Test Framework: Alcotest

Tests Passing: 42/42

API Endpoints

- GET /health - Health check
- POST /api/grammar/check - Check text for grammar issues
- POST /api/readability/score - Compute readability score (Flesch-like)
- POST /api/style/analyze - Analyze writing style
- POST /api/text/improve - Generate improved text
- POST /api/text/analyze - Full AI-powered analysis
- GET /api/stats - Usage statistics

Key Features

- Grammar checking: double spaces, capitalization, repeated words, missing punctuation
- Readability scoring using simplified Flesch formula
- Style analysis: passive voice detection, adverb counting, sentence variety
- Text improvement: auto-fix double spaces, capitalization, add ending punctuation
- Sentence variety measurement via standard deviation of sentence lengths

DevOps

Docker: Multi-stage build, OCaml opam switch, non-root user

CI/CD: GitHub Actions - opam setup, dune test + Docker build

Repository: <https://github.com/malbinjvc/grammar-forge>

Errors Encountered

- dune allow_empty error - package had no user-defined stanzas
- Missing lwt_ppx preprocessor for Dream's let%lwt async syntax
- libev not found on macOS - set C_INCLUDE_PATH and LIBRARY_PATH
- String.contains_s not in OCaml stdlib - used Str.search_forward
- Passive voice/adverb detection failing due to trailing punctuation on words
- Unused rec keyword on non-recursive function